Exercises: Constrained Optimization and the Implicit Function Theorem – Answer Key

Mathematical Foundations for Machine Learning

Implicit Function Theorem

Exercise 1.1 [Easy]
Consider the equation $F(x, y) = x^2 + y^2 - 1 = 0$ and the point $(x_0, y_0) = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$.

(a) Verify that $F(x_0, y_0) = 0$.

(b) Check the hypothesis of the Implicit Function Theorem: compute $\frac{\partial F}{\partial y}(x_0, y_0)$ and verify it is nonzero.

(c) Conclude that there exists a function $y = \varphi(x)$ defined near x_0 such that $F(x, \varphi(x)) = 0$. Find $\varphi'(x_0)$ by implicit differentiation.

Solution

(a) $F\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) = \frac{1}{2} + \frac{1}{2} - 1 = 0$. ✓

(b) $\frac{\partial F}{\partial y} = 2y$. At (x_0, y_0): $\frac{\partial F}{\partial y} = 2 \cdot \frac{1}{\sqrt{2}} = \sqrt{2} \neq 0$. The IFT hypothesis is satisfied.

(c) By the IFT, such φ exists. Differentiating $F(x, \varphi(x)) = 0$ implicitly:

$$\frac{\partial F}{\partial x} + \frac{\partial F}{\partial y} \cdot \varphi'(x) = 0 \implies \varphi'(x) = -\frac{\partial F/\partial x}{\partial F/\partial y} = -\frac{2x}{2y} = -\frac{x}{y}.$$

At (x_0, y_0): $\varphi'(x_0) = -\frac{1/\sqrt{2}}{1/\sqrt{2}} = -1$.

Exercise 1.2 [Medium]
Consider the system of equations $F : \mathbb{R}^3 \to \mathbb{R}$ defined by

$$F(x_1, x_2, x_3) = x_1^2 - x_1 x_3 + x_2^2 x_1 - x_2^2 = 0.$$

(a) Verify that $z = (1, 1, 1)$ is a solution.

(b) Can we locally solve for x_3 as a function of (x_1, x_2) near z? Justify using the Implicit Function Theorem.

(c) Compute $\frac{\partial x_3}{\partial x_1}$ and $\frac{\partial x_3}{\partial x_2}$ at the point $(1, 1)$ by implicit differentiation.

(d) At what points does the IFT fail to guarantee a local solution for x_3? Interpret geometrically.

Solution

(a) $F(1,1,1) = 1 - 1 + 1 - 1 = 0$. ✓
(b) We need $\frac{\partial F}{\partial x_3}(z) \neq 0$. We have $\frac{\partial F}{\partial x_3} = -x_1$. At $z = (1,1,1)$: $\frac{\partial F}{\partial x_3} = -1 \neq 0$. By the IFT, we can locally solve $x_3 = \varphi(x_1, x_2)$ near z.
(c) Differentiating $F(x_1, x_2, \varphi(x_1, x_2)) = 0$ with respect to x_1:

$$\frac{\partial F}{\partial x_1} + \frac{\partial F}{\partial x_3} \cdot \frac{\partial \varphi}{\partial x_1} = 0.$$

We have $\frac{\partial F}{\partial x_1} = 2x_1 - x_3 + x_2^2$. At z: $\frac{\partial F}{\partial x_1} = 2 - 1 + 1 = 2$.

$$\frac{\partial \varphi}{\partial x_1}(1,1) = -\frac{2}{-1} = 2.$$

Differentiating with respect to x_2:

$$\frac{\partial F}{\partial x_2} = 2x_2x_1 - 2x_2. \quad \text{At } z: \frac{\partial F}{\partial x_2} = 2 - 2 = 0.$$

$$\frac{\partial \varphi}{\partial x_2}(1,1) = -\frac{0}{-1} = 0.$$

(d) The IFT fails when $\frac{\partial F}{\partial x_3} = -x_1 = 0$, i.e., when $x_1 = 0$. At such points, the constraint becomes $-x_2^2 = 0 \Rightarrow x_2 = 0$, and x_3 can be arbitrary. Geometrically, the solution set has a "vertical" direction (parallel to the x_3-axis) at these points, so x_3 cannot be expressed as a function of (x_1, x_2).

Exercise 1.3 [Hard]
Let $F : \mathbb{R}^4 \to \mathbb{R}^2$ be defined by

$$F(x, y, u, v) = \begin{pmatrix} x^2 + y^2 - u^2 - v \\ xy - uv \end{pmatrix}.$$

Consider the point $z_0 = (1, 1, 1, 1)$.

(a) Verify that $F(z_0) = 0$.

(b) Can we locally solve for (u, v) as functions of (x, y) near z_0? Set up and verify the appropriate condition from the Implicit Function Theorem.

(c) Compute the Jacobian matrix $D_{(x,y)}(u, v)$ at $(1, 1)$ using the implicit differentiation formula:

$$D_y\varphi = -(D_yF)^{-1} \cdot D_xF,$$

where $\varphi(x, y) = (u(x, y), v(x, y))$.

Solution

(a) $F(1,1,1,1) = \begin{pmatrix} 1+1-1-1 \\ 1-1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$. ✓

(b) We need the Jacobian of F with respect to (u,v) to be invertible at z_0:

$$D_{(u,v)}F = \begin{pmatrix} \frac{\partial F_1}{\partial u} & \frac{\partial F_1}{\partial v} \\ \frac{\partial F_2}{\partial u} & \frac{\partial F_2}{\partial v} \end{pmatrix} = \begin{pmatrix} -2u & -1 \\ -v & -u \end{pmatrix}.$$

At z_0: $D_{(u,v)}F = \begin{pmatrix} -2 & -1 \\ -1 & -1 \end{pmatrix}$. The determinant is $(-2)(-1) - (-1)(-1) = 2 - 1 = 1 \neq 0$. The IFT applies.

(c) The Jacobian with respect to (x,y):

$$D_{(x,y)}F = \begin{pmatrix} 2x & 2y \\ y & x \end{pmatrix}. \quad \text{At } z_0 : \begin{pmatrix} 2 & 2 \\ 1 & 1 \end{pmatrix}.$$

We need $(D_{(u,v)}F)^{-1}$ at z_0:

$$\begin{pmatrix} -2 & -1 \\ -1 & -1 \end{pmatrix}^{-1} = \frac{1}{1}\begin{pmatrix} -1 & 1 \\ 1 & -2 \end{pmatrix} = \begin{pmatrix} -1 & 1 \\ 1 & -2 \end{pmatrix}.$$

Applying the formula:

$$D_{(x,y)}\varphi = -\begin{pmatrix} -1 & 1 \\ 1 & -2 \end{pmatrix}\begin{pmatrix} 2 & 2 \\ 1 & 1 \end{pmatrix} = -\begin{pmatrix} -2+1 & -2+1 \\ 2-2 & 2-2 \end{pmatrix} = -\begin{pmatrix} -1 & -1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}.$$

So $\frac{\partial u}{\partial x} = 1$, $\frac{\partial u}{\partial y} = 1$, $\frac{\partial v}{\partial x} = 0$, $\frac{\partial v}{\partial y} = 0$ at the point $(1,1)$.

Lagrange Multipliers — Setup and Computation

Exercise 2.1 [Easy]
Find the extrema of $f(x,y) = x + y$ subject to the constraint $x^2 + y^2 = 1$.

(a) Write down the Lagrangian $\mathcal{L}(x,y,\lambda)$.
(b) Compute the stationarity conditions $\nabla_{x,y}\mathcal{L} = 0$ and the constraint equation.
(c) Solve the system to find all critical points and classify them as maxima or minima.

Solution

(a) $\mathcal{L}(x, y, \lambda) = x + y - \lambda(x^2 + y^2 - 1)$.
(b) Stationarity:

$$\frac{\partial \mathcal{L}}{\partial x} = 1 - 2\lambda x = 0 \implies x = \frac{1}{2\lambda},$$
$$\frac{\partial \mathcal{L}}{\partial y} = 1 - 2\lambda y = 0 \implies y = \frac{1}{2\lambda}.$$

Constraint: $x^2 + y^2 = 1$.
(c) From stationarity, $x = y = \frac{1}{2\lambda}$. Substituting into the constraint:

$$2 \cdot \frac{1}{4\lambda^2} = 1 \implies \lambda^2 = \frac{1}{2} \implies \lambda = \pm\frac{1}{\sqrt{2}}.$$

$\lambda = \frac{1}{\sqrt{2}}$: $(x, y) = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$, $f = \sqrt{2}$ (maximum).
$\lambda = -\frac{1}{\sqrt{2}}$: $(x, y) = \left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$, $f = -\sqrt{2}$ (minimum).

Exercise 2.2 [Easy]
Find the point on the plane $2x + y + z = 5$ that is closest to the origin.

(a) Formulate this as a constrained optimization problem: state the objective function and the constraint.

(b) Write the Lagrangian and derive the stationarity conditions.

(c) Solve for the closest point.

Solution

(a) Minimize $f(x, y, z) = x^2 + y^2 + z^2$ subject to $g(x, y, z) = 2x + y + z - 5 = 0$.
(We minimize the squared distance; the minimizer is the same as for the Euclidean distance.)
(b) $\mathcal{L} = x^2 + y^2 + z^2 - \lambda(2x + y + z - 5)$.
Stationarity:

$$2x = 2\lambda \implies x = \lambda, \quad 2y = \lambda \implies y = \frac{\lambda}{2}, \quad 2z = \lambda \implies z = \frac{\lambda}{2}.$$

(c) Substituting into $2x + y + z = 5$:

$$2\lambda + \frac{\lambda}{2} + \frac{\lambda}{2} = 5 \implies 3\lambda = 5 \implies \lambda = \frac{5}{3}.$$

The closest point is $\left(\frac{5}{3}, \frac{5}{6}, \frac{5}{6}\right)$.

Exercise 2.3 [Medium]
Consider the curve $y = 1.5 - x^2$ in $\mathbb{R}^2$.

(a) Formulate the problem of finding the point(s) on this curve closest to the origin as a constrained optimization problem using the squared Euclidean distance.

(b) Write down the Lagrangian. Derive the stationarity conditions.

(c) Solve the system. You should find that there are two cases depending on the value of λ. Find all constrained minimizers.

(d) State the necessary and sufficient conditions for a point to be a constrained minimizer in general.

Solution

(a) Minimize $f(x, y) = x^2 + y^2$ subject to $g(x, y) = y - 1.5 + x^2 = 0$.
(b) $\mathcal{L}(x, y, \lambda) = x^2 + y^2 - \lambda(y - 1.5 + x^2)$.
Stationarity:

$$\frac{\partial \mathcal{L}}{\partial x} = 2x - 2\lambda x = 0 \implies 2x(1 - \lambda) = 0,$$

$$\frac{\partial \mathcal{L}}{\partial y} = 2y - \lambda = 0 \implies y = \frac{\lambda}{2}.$$

Constraint: $y = 1.5 - x^2$.
(c) From the first equation, either $x = 0$ or $\lambda = 1$.
Case 1: $x = 0$. Then $y = 1.5$ and $\lambda = 2y = 3$. Critical point: $(0, 1.5)$ with $f = 2.25$.
Case 2: $\lambda = 1$. Then $y = 1/2$. From the constraint: $1/2 = 1.5 - x^2 \implies x^2 = 1 \implies x = \pm 1$. Critical points: $(1, 0.5)$ and $(-1, 0.5)$ with $f = 1.25$.
The constrained minimizers are $(\pm 1, 0.5)$ with squared distance 1.25.
(d) Necessary condition (Theorem of Lagrange): If c is a local extremum of f on $N = \{x : g(x) = 0\}$ and $\nabla g(c) \neq 0$, then $\exists\, \lambda$ such that $\nabla f(c) = \lambda \nabla g(c)$.
Sufficient condition: Let $h = f - \lambda g$. If $\nabla h(c) = 0$ and $\langle v, Hh(c)\, v\rangle > 0$ for all $v \neq 0$ satisfying $\langle v, \nabla g(c)\rangle = 0$, then c is a local constrained minimum.

Exercise 2.4 [Medium]
Find the closest point to $(0, 0)$ in the set

$$M = \{x \in \mathbb{R}^2 : x_1 + x_2 \geq 4,\ 2x_1 + x_2 \geq 5\}.$$

(a) Formulate as an optimization problem with inequality constraints. Write the objective and constraints in the standard form $g_i(x) \leq 0$.
(b) Write the Lagrangian and state all four KKT conditions.
(c) Solve the KKT system. Use complementary slackness to enumerate cases.

Solution

(a) Minimize $f(x) = x_1^2 + x_2^2$ subject to:

$$g_1(x) = 4 - x_1 - x_2 \leq 0, \qquad g_2(x) = 5 - 2x_1 - x_2 \leq 0.$$

(b) $\mathcal{L} = x_1^2 + x_2^2 + \lambda_1(4 - x_1 - x_2) + \lambda_2(5 - 2x_1 - x_2)$.
KKT conditions:

1. **Stationarity:** $2x_1 - \lambda_1 - 2\lambda_2 = 0, \quad 2x_2 - \lambda_1 - \lambda_2 = 0.$
2. **Primal feasibility:** $x_1 + x_2 \geq 4, \quad 2x_1 + x_2 \geq 5.$
3. **Dual feasibility:** $\lambda_1, \lambda_2 \geq 0.$
4. **Complementary slackness:** $\lambda_1(4 - x_1 - x_2) = 0, \quad \lambda_2(5 - 2x_1 - x_2) = 0.$

(c) We enumerate cases via complementary slackness.
Case 1: $\lambda_1 > 0, \lambda_2 = 0$**.** Both constraints active for g_1: $x_1 + x_2 = 4$. Stationarity gives $2x_1 = \lambda_1$, $2x_2 = \lambda_1$, so $x_1 = x_2$. Then $2x_1 = 4 \implies x_1 = x_2 = 2$, $\lambda_1 = 4$. Check g_2: $4 + 2 - 5 = 1 \geq 0$. ✓. Feasible. $f = 8$.
Case 2: $\lambda_1 = 0, \lambda_2 > 0$**.** Active: $2x_1 + x_2 = 5$. Stationarity: $2x_1 = 2\lambda_2$, $2x_2 = \lambda_2$, so $x_1 = \lambda_2$, $x_2 = \lambda_2/2$, thus $x_1 = 2x_2$. Then $4x_2 + x_2 = 5 \implies x_2 = 1$, $x_1 = 2$. Check g_1: $2 + 1 = 3 < 4$. Not feasible. Discard.
Case 3: $\lambda_1 > 0, \lambda_2 > 0$**.** Both active: $x_1 + x_2 = 4$ and $2x_1 + x_2 = 5$. Subtracting: $x_1 = 1$, $x_2 = 3$. Stationarity: $2 = \lambda_1 + 2\lambda_2$, $6 = \lambda_1 + \lambda_2$. Subtracting: $\lambda_2 = -4 < 0$. Violates dual feasibility. Discard.
The unique solution is $\boxed{(x_1, x_2) = (2, 2)}$ with $f = 8$.

Exercise 2.5 [Medium]
Consider the entropy maximization problem on the probability simplex:

$$\max_{x \in \mathbb{R}^n} \left(-\sum_{i=1}^{n} x_i \log x_i \right) \quad \text{s.t.} \quad \sum_{i=1}^{n} x_i = 1, \quad x_i \geq 0.$$

(a) Write the Lagrangian incorporating both the equality and inequality constraints.
(b) Derive the stationarity condition with respect to x_i.
(c) Using complementary slackness, argue that $x_i > 0$ for all i at the optimum. Then solve for x_i.
(d) What is the maximum entropy value?

Solution

(a) Converting to minimization of $\sum x_i \log x_i$:

$$\mathcal{L} = \sum_{i=1}^{n} x_i \log x_i + \mu \left(\sum_{i=1}^{n} x_i - 1 \right) + \sum_{i=1}^{n} \nu_i(-x_i),$$

with $\nu_i \geq 0$.
(b) Stationarity with respect to x_i:

$$\frac{\partial \mathcal{L}}{\partial x_i} = \log x_i + 1 + \mu - \nu_i = 0 \implies \log x_i = -(1+\mu) + \nu_i.$$

(c) By complementary slackness: $\nu_i \cdot (-x_i) = 0$.
Suppose $x_i = 0$ for some i. Then $\log x_i \to -\infty$, but $-(1+\mu) + \nu_i$ is finite, giving a contradiction. So $x_i > 0$ for all i, which forces $\nu_i = 0$ for all i.
With $\nu_i = 0$: $\log x_i = -(1+\mu)$ for all i, so all x_i are equal. From $\sum x_i = 1$: $x_i = \frac{1}{n}$ for all i.
(d) The maximum entropy is:

$$-\sum_{i=1}^{n} \frac{1}{n} \log \frac{1}{n} = -n \cdot \frac{1}{n} \cdot (-\log n) = \log n.$$

Lagrange Multipliers — Proofs and Theory

Exercise 3.1 [Easy]
Let $g : \mathbb{R}^n \to \mathbb{R}$ be differentiable and $N = \{x : g(x) = 0\}$. Suppose $c \in N$ is a local minimum of a differentiable function f on N, and $\nabla g(c) \neq 0$.

(a) State the necessary condition (Lagrange condition) that must hold at c.

(b) Give a geometric interpretation: what does $\nabla f(c) = \lambda \nabla g(c)$ mean about the gradients at the constrained optimum?

(c) Why is the condition $\nabla g(c) \neq 0$ (constraint qualification) needed?

Solution

(a) There exists $\lambda \in \mathbb{R}$ such that $\nabla f(c) = \lambda \nabla g(c)$.
(b) The gradient $\nabla f(c)$ is parallel to $\nabla g(c)$ at the optimum. Since $\nabla g(c)$ is normal to the constraint surface N, this means that $\nabla f(c)$ is also normal to N at c. Equivalently, the component of ∇f tangent to the constraint surface is zero: there is no direction along N in which f decreases, which is necessary for a constrained extremum.
(c) If $\nabla g(c) = 0$, the constraint surface may have a singularity (e.g., a cusp or crossing) at c. The Implicit Function Theorem cannot be applied to locally parametrize N, so the tangent space to N at c is not well-defined. Without a well-defined tangent space, the Lagrange condition need not hold. As a counterexample, consider $f(x, y) = x$ and $g(x, y) = x^3 - y^2$: the origin satisfies $g = 0$ and is a local minimum of f on N, but $\nabla g(0, 0) = (0, 0)$ and $\nabla f(0, 0) = (1, 0) \neq \lambda(0, 0)$ for any λ.

Exercise 3.2 [Medium]
Proving the Lagrange necessary condition in $\mathbb{R}^2$.

Let $f, g : \mathbb{R}^2 \to \mathbb{R}$ be C^1 functions, $N = \{(x, y) : g(x, y) = 0\}$, and suppose $c = (a, b) \in N$ is a local minimum of f on N with $\frac{\partial g}{\partial y}(c) \neq 0$.

(a) By the Implicit Function Theorem, there exists φ with $g(x, \varphi(x)) = 0$ near a. Define $h(x) = f(x, \varphi(x))$. Explain why $h'(a) = 0$.

(b) Compute $h'(a)$ using the chain rule and implicit differentiation (express $\varphi'(a)$ in terms of partial derivatives of g).

(c) Use parts (a) and (b) to show that $\nabla f(c) = \lambda \nabla g(c)$ for some $\lambda \in \mathbb{R}$.

Solution

(a) Since c is a local minimum of f on N and $(x, \varphi(x))$ parametrizes N locally, $h(x) = f(x, \varphi(x))$ is a real-valued function with a local minimum at $x = a$. By the standard necessary condition for unconstrained extrema, $h'(a) = 0$.

(b) By the chain rule:

$$h'(x) = \frac{\partial f}{\partial x}(x, \varphi(x)) + \frac{\partial f}{\partial y}(x, \varphi(x)) \cdot \varphi'(x).$$

From $g(x, \varphi(x)) = 0$, implicit differentiation gives:

$$\frac{\partial g}{\partial x} + \frac{\partial g}{\partial y} \cdot \varphi'(x) = 0 \implies \varphi'(a) = -\frac{\partial g/\partial x(c)}{\partial g/\partial y(c)}.$$

So: $h'(a) = \frac{\partial f}{\partial x}(c) - \frac{\partial f}{\partial y}(c) \cdot \frac{\partial g/\partial x(c)}{\partial g/\partial y(c)}$.

(c) Setting $h'(a) = 0$:

$$\frac{\partial f}{\partial x}(c) \cdot \frac{\partial g}{\partial y}(c) = \frac{\partial f}{\partial y}(c) \cdot \frac{\partial g}{\partial x}(c).$$

Define $\lambda = \frac{\partial f/\partial y(c)}{\partial g/\partial y(c)}$ (well-defined since $\frac{\partial g}{\partial y}(c) \neq 0$). Then:

$$\frac{\partial f}{\partial y}(c) = \lambda \frac{\partial g}{\partial y}(c) \quad \text{(by definition of } \lambda).$$

Substituting back: $\frac{\partial f}{\partial x}(c) \cdot \frac{\partial g}{\partial y}(c) = \lambda \frac{\partial g}{\partial x}(c) \cdot \frac{\partial g}{\partial y}(c)$. Since $\frac{\partial g}{\partial y}(c) \neq 0$, we can divide:

$$\frac{\partial f}{\partial x}(c) = \lambda \frac{\partial g}{\partial x}(c).$$

Thus $\nabla f(c) = \lambda \nabla g(c)$. ■

Exercise 3.3 [Hard]

Sufficient condition for a constrained minimum.

Let $f, g : \mathbb{R}^n \to \mathbb{R}$ be C^2, $N = \{x : g(x) = 0\}$, and suppose at $c \in N$ there exists λ such that the Lagrangian $h = f - \lambda g$ satisfies $\nabla h(c) = 0$.

Define the **constrained tangent space** $T_c = \{v \in \mathbb{R}^n : \langle v, \nabla g(c) \rangle = 0\}$.

(a) State the sufficient condition for c to be a strict local constrained minimum in terms of the Hessian of h restricted to T_c.

(b) *Application.* Consider $f(x, y) = x^2 + 2y^2$ subject to $g(x, y) = x + y - 1 = 0$. Find the critical point using Lagrange multipliers. Then verify the sufficient condition to confirm it is a local minimum.

(c) *Application.* Consider maximizing $f(x, y, z) = xyz$ on the sphere $x^2 + y^2 + z^2 = 1$. Find the critical points in the first octant $(x, y, z > 0)$. Verify using the bordered Hessian or the restricted Hessian that this is indeed a local maximum.

Solution

(a) c is a strict local constrained minimum if $\langle v, Hh(c)\, v\rangle > 0$ for all $v \neq 0$ with $v \in T_c$, i.e., the Hessian of $h = f - \lambda g$ is positive definite when restricted to the tangent space of the constraint.
(b) Lagrangian: $\mathcal{L} = x^2 + 2y^2 - \lambda(x + y - 1)$.
Stationarity: $2x = \lambda$, $4y = \lambda$. So $x = \lambda/2$, $y = \lambda/4$. Constraint: $\lambda/2 + \lambda/4 = 1 \implies \lambda = 4/3$. Critical point: $c = (2/3, 1/3)$.
$h = f - \lambda g = x^2 + 2y^2 - \frac{4}{3}(x + y - 1)$. Hessian:

$$Hh = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix}.$$

The tangent space $T_c = \{v : v_1 + v_2 = 0\}$, so $v = t(-1, 1)^T$ for $t \neq 0$.

$$v^T Hh\, v = t^2(-1, 1) \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix} \begin{pmatrix} -1 \\ 1 \end{pmatrix} = t^2(2 + 4) = 6t^2 > 0.$$

Sufficient condition is satisfied: $c = (2/3, 1/3)$ is a strict local constrained minimum.
(c) Constraint: $g(x, y, z) = x^2 + y^2 + z^2 - 1 = 0$.
Lagrangian: $\mathcal{L} = xyz - \lambda(x^2 + y^2 + z^2 - 1)$.
Stationarity:

$$yz = 2\lambda x, \quad xz = 2\lambda y, \quad xy = 2\lambda z.$$

From the first two: $yz/x = xz/y \implies y^2 = x^2$. Since $x, y > 0$: $x = y$. Similarly $y = z$. So $x = y = z$. From the constraint: $3x^2 = 1 \implies x = 1/\sqrt{3}$. Then $\lambda = \frac{yz}{2x} = \frac{1/3}{2/\sqrt{3}} = \frac{1}{2\sqrt{3}}$.
Critical point: $c = (1/\sqrt{3}, 1/\sqrt{3}, 1/\sqrt{3})$ with $f(c) = 1/(3\sqrt{3})$.
$h = xyz - \lambda(x^2 + y^2 + z^2 - 1)$. The Hessian of h:

$$Hh = \begin{pmatrix} -2\lambda & z & y \\ z & -2\lambda & x \\ y & x & -2\lambda \end{pmatrix} = \begin{pmatrix} -1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \\ 1/\sqrt{3} & -1/\sqrt{3} & 1/\sqrt{3} \\ 1/\sqrt{3} & 1/\sqrt{3} & -1/\sqrt{3} \end{pmatrix}.$$

$T_c = \{v : 2xv_1 + 2yv_2 + 2zv_3 = 0\} = \{v : v_1 + v_2 + v_3 = 0\}$.
For $v \in T_c$ with $v_3 = -v_1 - v_2$:

$$v^T Hh\, v = \frac{1}{\sqrt{3}}\big[-v_1^2 - v_2^2 - (v_1 + v_2)^2 + 2v_1v_2 - 2v_1(v_1 + v_2) - 2v_2(v_1 + v_2)\big].$$

Simplifying: $= \frac{1}{\sqrt{3}}(-2v_1^2 - 2v_2^2 - 2v_1v_2) \cdot 1 = \frac{-2}{\sqrt{3}}(v_1^2 + v_1v_2 + v_2^2)$.
Since $v_1^2 + v_1v_2 + v_2^2 = (v_1 + v_2/2)^2 + 3v_2^2/4 > 0$ for $v \neq 0$, we have $v^T Hh\, v < 0$ on T_c. This confirms c is a strict local constrained **maximum**.

Lagrange Multipliers with Inequality Constraints (KKT)

Exercise 4.1 [Easy]
State all four Karush–Kuhn–Tucker (KKT) conditions for the problem

$$\min_x f(x) \quad \text{s.t.} \quad g_i(x) \leq 0,\ i = 1, \ldots, m, \quad h_j(x) = 0,\ j = 1, \ldots, p.$$

Briefly explain the role of each condition.

Solution

The KKT conditions for the Lagrangian $\mathcal{L}(x, \lambda, \mu) = f(x) + \sum_{i=1}^{m} \lambda_i g_i(x) + \sum_{j=1}^{p} \mu_j h_j(x)$ are:
1. Stationarity: $\nabla_x \mathcal{L} = \nabla f(x^*) + \sum_{i=1}^{m} \lambda_i \nabla g_i(x^*) + \sum_{j=1}^{p} \mu_j \nabla h_j(x^*) = 0$.
Role: The gradient of the objective is a linear combination of the constraint gradients at the optimum — no feasible descent direction exists.
2. Primal feasibility: $g_i(x^*) \leq 0$ for all i, and $h_j(x^*) = 0$ for all j.
Role: The solution must satisfy all constraints.
3. Dual feasibility: $\lambda_i \geq 0$ for all i.
Role: The multipliers for inequality constraints are non-negative (they "push" the objective only in the correct direction).
4. Complementary slackness: $\lambda_i g_i(x^*) = 0$ for all i.
Role: If a constraint is not active ($g_i(x^*) < 0$), its multiplier must be zero — inactive constraints play no role at the optimum.

Exercise 4.2 [Medium]
Minimize $f(x_1, x_2) = (x_1 - 3)^2 + (x_2 - 2)^2$ subject to

$$x_1 + x_2 \leq 4, \quad x_1 \geq 0, \quad x_2 \geq 0.$$

(a) Rewrite the constraints in standard form $g_i(x) \leq 0$.
(b) Write the Lagrangian and the full set of KKT conditions.
(c) Solve the system. Enumerate cases using complementary slackness and find the optimal solution.

Solution

(a) $g_1(x) = x_1 + x_2 - 4 \leq 0$, $g_2(x) = -x_1 \leq 0$, $g_3(x) = -x_2 \leq 0$.
(b) $\mathcal{L} = (x_1 - 3)^2 + (x_2 - 2)^2 + \lambda_1(x_1 + x_2 - 4) + \lambda_2(-x_1) + \lambda_3(-x_2)$.
KKT conditions:

- Stationarity: $2(x_1 - 3) + \lambda_1 - \lambda_2 = 0, \quad 2(x_2 - 2) + \lambda_1 - \lambda_3 = 0$.
- Primal feasibility: $x_1 + x_2 \leq 4$, $x_1 \geq 0$, $x_2 \geq 0$.
- Dual feasibility: $\lambda_1, \lambda_2, \lambda_3 \geq 0$.
- Complementary slackness: $\lambda_1(x_1 + x_2 - 4) = 0$, $\lambda_2 x_1 = 0$, $\lambda_3 x_2 = 0$.

(c) First try the unconstrained minimizer: $(3, 2)$ with $f = 0$. Check feasibility: $3 + 2 = 5 > 4$. Not feasible.
Since the closest point to the unconstrained optimum will be on the boundary, try $\lambda_1 > 0$ (constraint g_1 active), $\lambda_2 = \lambda_3 = 0$.
Active: $x_1 + x_2 = 4$.
Stationarity: $2(x_1 - 3) + \lambda_1 = 0$ and $2(x_2 - 2) + \lambda_1 = 0$.
Subtracting: $2(x_1 - 3) = 2(x_2 - 2) \implies x_1 - x_2 = 1$. Combined with $x_1 + x_2 = 4$: $x_1 = 5/2$, $x_2 = 3/2$.
$\lambda_1 = -2(5/2 - 3) = 1 > 0$. Check $x_1, x_2 > 0$. ✓.
Optimal solution: $(x_1, x_2) = (5/2, 3/2)$ with $f = (5/2 - 3)^2 + (3/2 - 2)^2 = 1/4 + 1/4 = 1/2$.

Exercise 4.3 [Hard]

Consider the KL-divergence minimization problem from the 2026 exam:

$$\min_{x\in\mathbb{R}^n} \sum_{i=1}^n x_i \left(\log x_i - \log p_i\right) \quad \text{s.t.} \quad \sum_{i=1}^n x_i = 1, \quad x_i \geq 0,$$

where $p_i > 0$ are given with $\sum_i p_i = 1$.

(a) Is the objective function convex? Justify.

(b) Is the objective function **strongly** convex? Prove or disprove.

(c) Assuming strong duality holds, write down the Lagrangian, the dual function (minimizing over x), and solve the dual problem.

(d) What is the optimal solution x^*? Interpret the result in terms of information theory.

Solution

(a) Write $\phi(x) = \sum_i x_i(\log x_i - \log p_i) = \sum_i \phi_i(x_i)$ where $\phi_i(t) = t\log t - t\log p_i$.
$\phi_i''(t) = 1/t > 0$ for $t > 0$. Since ϕ is a sum of convex functions, it is convex. ✓
(b) Strong convexity requires $H\phi(x) \succeq mI$ for some $m > 0$ for all feasible x. The Hessian is $H\phi(x) = \operatorname{diag}(1/x_1, \ldots, 1/x_n)$. As $x_i \to 0^+$, the eigenvalue $1/x_i \to \infty$, but no *uniform* lower bound $m > 0$ is valid on the open feasible set (near the boundary). More precisely, on the compact feasible set $\{x : \sum x_i = 1, x_i \geq 0\}$, as $x_i \to 0^+$, $1/x_j$ for other coordinates is bounded but we need *all* eigenvalues $\geq m$. Since the feasible region includes points where some x_j is close to 1 and the minimum eigenvalue is $\min_i 1/x_i = 1/\max_i x_i$, which can approach 1, strong convexity holds on the interior with $m = 1/\max x_i \leq 1$ but there is no *uniform* constant on the entire domain $\mathbb{R}^n_{>0}$.
More directly: ϕ is **not** strongly convex on $\mathbb{R}^n_{>0}$ because for x with some $x_i \to \infty$, $1/x_i \to 0$, so $H\phi \not\succeq mI$ globally. (However, restricted to the simplex it is well-behaved.)
(c) Lagrangian (using μ for equality, $\nu_i \geq 0$ for $x_i \geq 0$):

$$\mathcal{L}(x, \mu, \nu) = \sum_i x_i(\log x_i - \log p_i) + \mu\left(\sum_i x_i - 1\right) - \sum_i \nu_i x_i.$$

Stationarity: $\frac{\partial \mathcal{L}}{\partial x_i} = \log x_i + 1 - \log p_i + \mu - \nu_i = 0$.
By the same argument as Exercise 2.5, at the optimum $x_i > 0$ so $\nu_i = 0$:

$$\log x_i = \log p_i - 1 - \mu \implies x_i = p_i \cdot e^{-(1+\mu)}.$$

The dual function: substituting back,

$$g(\mu) = \sum_i p_i e^{-(1+\mu)}(-(1+\mu) - \log p_i + \log p_i) + \mu(e^{-(1+\mu)} \sum_i p_i - 1).$$

Simplifying with $\sum p_i = 1$:

$$g(\mu) = -(1+\mu)e^{-(1+\mu)} + \mu(e^{-(1+\mu)} - 1) = -e^{-(1+\mu)} - \mu.$$

Maximizing: $g'(\mu) = e^{-(1+\mu)} - 1 = 0 \implies e^{-(1+\mu)} = 1 \implies \mu = -1$.
Then $x_i = p_i \cdot e^0 = p_i$.
(d) The optimal solution is $x^* = p$, i.e., $x_i^* = p_i$ for all i.
The minimum value of the KL divergence $D_{\mathrm{KL}}(x\|p)$ is 0, achieved when $x = p$. This is the fundamental property that KL divergence is non-negative and equals zero if and only if the two distributions are identical.

Mixed Challenging Exercises

Exercise 5.1 [Hard]
Spectral optimization via Lagrange multipliers.

Let $A \in \mathbb{R}^{n\times n}$ be a symmetric matrix with eigenvalues $\lambda_1 \le \lambda_2 \le \cdots \le \lambda_n$.

(a) Formulate the problem $\max_{\|x\|_2=1} x^T Ax$ as a constrained optimization problem and apply the method of Lagrange multipliers. Show that the critical points satisfy $Ax = \mu x$ for some multiplier μ, i.e., they are eigenvectors.

(b) Conclude that the maximum of the Rayleigh quotient $R_A(x) = \frac{x^T Ax}{x^T x}$ equals λ_n.

(c) Now consider the 2026 exam-style problem: $A = aa^T + bb^T$ with $a = u + v$, $b = \frac{3}{4}u - 3v$, where u, v are orthonormal. Find $\max_{\|x\|=1} \|Ax\|_2 - \langle x, Ax\rangle$.

(*Hint: First compute the eigenvalues of A, then express the objective in spectral coordinates.*)

Solution

(a) Maximize $f(x) = x^T Ax$ subject to $g(x) = x^T x - 1 = 0$.
Lagrangian: $\mathcal{L} = x^T Ax - \mu(x^T x - 1)$.
Stationarity: $\nabla_x \mathcal{L} = 2Ax - 2\mu x = 0 \implies Ax = \mu x$.
Thus every critical point is an eigenvector with μ being the corresponding eigenvalue.
(b) At a critical point $x = v_i$ (eigenvector), $f(v_i) = v_i^T Av_i = \lambda_i$. The maximum over all critical points is λ_n.
Since f is continuous on the compact set $\{\|x\| = 1\}$, a maximum exists. It must occur at a critical point, so $\max_{\|x\|=1} x^T Ax = \lambda_n$.
(c) First compute eigenvalues. A acts on $\text{span}(u, v)$ with everything in $\text{span}(u, v)^\perp$ having eigenvalue 0.
In the basis $\{u, v\}$: $a = u + v$ so aa^T restricted to $\text{span}(u, v)$ has matrix $\begin{pmatrix}1\\1\end{pmatrix}(1, 1) = \begin{pmatrix}1 & 1\\1 & 1\end{pmatrix}$.
$b = \frac{3}{4}u - 3v$ so bb^T restricted has matrix $\begin{pmatrix}9/16 & -9/4\\-9/4 & 9\end{pmatrix}$.
$A|_{\text{span}} = \begin{pmatrix}25/16 & -5/4\\-5/4 & 10\end{pmatrix}$.
Eigenvalues: $\text{tr} = 25/16 + 10 = 185/16$, $\det = 250/16 - 25/16 = 225/16$.
$\lambda = \frac{185/16 \pm \sqrt{(185/16)^2 - 4 \cdot 225/16}}{2} = \frac{185 \pm \sqrt{185^2 - 4 \cdot 225 \cdot 16}}{32}$.
$185^2 = 34225$, $4 \cdot 225 \cdot 16 = 14400$. $\sqrt{34225 - 14400} = \sqrt{19825} = 5\sqrt{793}$.
$\lambda_{1,2} = \frac{185 \pm 5\sqrt{793}}{32}$.
For x on the unit sphere written in eigenbasis: $x = \sum c_i e_i$ with $\sum c_i^2 = 1$,

$$\|Ax\|_2 = \sqrt{\sum \lambda_i^2 c_i^2}, \quad \langle x, Ax\rangle = \sum \lambda_i c_i^2.$$

The objective $\|Ax\|_2 - \langle x, Ax\rangle$ is maximized by testing eigenvectors (and the zero-eigenvalue directions).
At eigenvector e_i with eigenvalue $\lambda_i \geq 0$: objective $= |\lambda_i| - \lambda_i = 0$ (for PSD matrix).
At eigenvector with $\lambda_i = 0$: objective $= 0 - 0 = 0$.
Since $A = aa^T + bb^T$ is PSD, all eigenvalues ≥ 0, and $\|Ax\|_2 \geq \langle x, Ax\rangle \geq 0$ always. The objective is 0 at every eigenvector. By careful analysis (or numerically), the maximum is achieved at a non-eigenvector direction mixing different eigenspaces. However, by Cauchy–Schwarz $\|Ax\|_2 \geq x^T Ax/\|x\| = x^T Ax$ with equality iff $Ax \parallel x$, i.e., x is an eigenvector. So the objective $\|Ax\| - x^T Ax$ is maximized when Ax is as far from parallel to x as possible.
For $x = \alpha e_1 + \beta e_2$ ($\alpha^2 + \beta^2 = 1$, restricting to the nontrivial subspace): $\|Ax\| = \sqrt{\lambda_1^2 \alpha^2 + \lambda_2^2 \beta^2}$ and $\langle x, Ax\rangle = \lambda_1 \alpha^2 + \lambda_2 \beta^2$. The maximum of the difference can be found by calculus over $\alpha^2 \in [0, 1]$, yielding the answer in terms of λ_1 and λ_2.

Exercise 5.2 [Hard]
From constrained optimization to MLE.

Consider N data points $\{(x_n, y_n)\}_{n=1}^N$ with $x_n \in \mathbb{R}^D$, $y_n \in \mathbb{R}$, arising from $y = x^T\theta + \varepsilon$ where $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ i.i.d.

(a) Write the conditional probability $P(Y|X; \theta)$ using the i.i.d. assumption.

(b) Show that the maximum likelihood estimate is $\hat{\theta} = (X^T X)^{-1} X^T Y$.

(c) Now suppose we add a regularization constraint $\|\theta\|_2^2 \leq r^2$ (Ridge-style). Formulate the constrained MLE as a Lagrangian problem and use KKT conditions to show that the solution

satisfies

$$\hat{\theta}_{\text{ridge}} = (X^T X + \lambda I)^{-1} X^T Y$$

for some $\lambda \geq 0$.

(d) Interpret the role of λ: what happens when $\lambda = 0$ vs. $\lambda \to \infty$?

Solution

(a) Each $y_n | x_n \sim \mathcal{N}(x_n^T \theta, \sigma^2)$. By independence:

$$P(Y|X;\theta) = \prod_{n=1}^{N} \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y_n - x_n^T\theta)^2}{2\sigma^2}\right).$$

(b) Log-likelihood:

$$\ell(\theta) = -\frac{N}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{n=1}^{N}(y_n - x_n^T\theta)^2 = -\frac{1}{2\sigma^2}\|Y - X\theta\|^2 + \text{const.}$$

Maximizing $\ell \iff$ minimizing $\|Y - X\theta\|^2$.
$\nabla_\theta \|Y - X\theta\|^2 = -2X^T(Y - X\theta) = 0 \implies X^T X\theta = X^T Y \implies \hat{\theta} = (X^T X)^{-1} X^T Y$.
(c) We minimize $f(\theta) = \frac{1}{2}\|Y - X\theta\|^2$ subject to $g(\theta) = \|\theta\|^2 - r^2 \leq 0$.
Lagrangian: $\mathcal{L} = \frac{1}{2}\|Y - X\theta\|^2 + \lambda(\|\theta\|^2 - r^2)$.
KKT stationarity:

$$\nabla_\theta \mathcal{L} = -X^T(Y - X\theta) + 2\lambda\theta = 0 \implies (X^T X + 2\lambda I)\theta = X^T Y.$$

Renaming $2\lambda \to \lambda$ (absorbing the factor): $\hat{\theta}_{\text{ridge}} = (X^T X + \lambda I)^{-1} X^T Y$ with $\lambda \geq 0$.
Complementary slackness: $\lambda(\|\theta\|^2 - r^2) = 0$. If the constraint is inactive ($\|\hat{\theta}\| < r$), then $\lambda = 0$ and we recover ordinary least squares.
(d) $\lambda = 0$: No regularization, we recover the OLS estimate $\hat{\theta} = (X^T X)^{-1} X^T Y$. The constraint $\|\theta\| \leq r$ is not active.
$\lambda \to \infty$: $(X^T X + \lambda I)^{-1} \approx \lambda^{-1} I$, so $\hat{\theta} \to 0$. Maximum regularization shrinks all parameters to zero, underfitting the data. The constraint ball $r \to 0$.
The parameter λ trades off between fitting the data (small λ) and keeping the model simple (large λ).

Exercise 5.3 [Hard]
Lagrange multipliers meet the Implicit Function Theorem.

Let $f, g : \mathbb{R}^3 \to \mathbb{R}$ be C^2 functions. Define the Lagrangian system $F : \mathbb{R}^4 \to \mathbb{R}^3$ by

$$F(x_1, x_2, x_3, \lambda) = \begin{pmatrix} \nabla f(x) - \lambda \nabla g(x) \\ g(x) \end{pmatrix} = \begin{pmatrix} \partial_1 f - \lambda \partial_1 g \\ \partial_2 f - \lambda \partial_2 g \\ \partial_3 f - \lambda \partial_3 g \\ g(x) \end{pmatrix} \in \mathbb{R}^4.$$

(Note: F maps $\mathbb{R}^4 \to \mathbb{R}^4$.)

Suppose (x^*, λ^*) is a solution of $F = 0$.

(a) Under what conditions on f and g can we apply the Implicit Function Theorem to conclude that the solution (x^*, λ^*) is locally unique (isolated)?

(b) Compute the Jacobian DF explicitly in terms of the Hessians of f and g and the gradient ∇g.

(c) *Specific case.* Let $f(x) = x_1^2 + x_2^2 + x_3^2$ and $g(x) = x_1 + x_2 + x_3 - 3 = 0$. Find the constrained minimum using Lagrange multipliers and verify that the Jacobian DF is invertible at the solution.

Solution

(a) The system $F(x, \lambda) = 0$ is a system of 4 equations in 4 unknowns (since $F : \mathbb{R}^4 \to \mathbb{R}^4$). By the Implicit Function Theorem (or, more precisely, the Inverse Function Theorem), the solution is locally unique if $\det(DF(x^*, \lambda^*)) \neq 0$.
This requires:

- The bordered Hessian of the Lagrangian is nonsingular, which combines the second-order information of $f - \lambda g$ with the constraint gradient ∇g.
- In particular, a sufficient condition is that $\nabla g(x^*) \neq 0$ and the Hessian of $f - \lambda^* g$ restricted to the tangent space of the constraint is nondegenerate (no zero eigenvalues on T_{x^*}).

(b)

$$DF(x, \lambda) = \begin{pmatrix} Hf(x) - \lambda Hg(x) & -\nabla g(x) \\ (\nabla g(x))^T & 0 \end{pmatrix} \in \mathbb{R}^{4\times 4},$$

where $Hf, Hg \in \mathbb{R}^{3\times 3}$ are the Hessians and $\nabla g \in \mathbb{R}^3$ is a column vector.
This is exactly the **bordered Hessian** matrix used in constrained optimization theory.
(c) Lagrangian: $\mathcal{L} = x_1^2 + x_2^2 + x_3^2 - \lambda(x_1 + x_2 + x_3 - 3)$.
Stationarity: $2x_i = \lambda$ for $i = 1, 2, 3$, so $x_1 = x_2 = x_3 = \lambda/2$.
Constraint: $3\lambda/2 = 3 \implies \lambda = 2$. Solution: $x^* = (1, 1, 1)$, $\lambda^* = 2$.
Now $Hf = 2I_3$, $Hg = 0$ (since g is linear), $\nabla g = (1, 1, 1)^T$.

$$DF = \begin{pmatrix} 2I_3 & -(1,1,1)^T \\ (1,1,1) & 0 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 & -1 \\ 0 & 2 & 0 & -1 \\ 0 & 0 & 2 & -1 \\ 1 & 1 & 1 & 0 \end{pmatrix}.$$

Expanding the determinant along the last row:

$$\det = 1 \cdot \det \begin{pmatrix} 0 & 0 & -1 \\ 2 & 0 & -1 \\ 0 & 2 & -1 \end{pmatrix} \cdot (-1) + 1 \cdot \det \begin{pmatrix} 2 & 0 & -1 \\ 0 & 0 & -1 \\ 0 & 2 & -1 \end{pmatrix} \cdot 1 + 1 \cdot \det \begin{pmatrix} 2 & 0 & -1 \\ 0 & 2 & -1 \\ 0 & 0 & -1 \end{pmatrix} \cdot (-1).$$

Computing: $\det_1 = 0\cdot(0\cdot(-1) - (-1)\cdot 2) - 0 + (-1)(4) = -4$. $\det_2 = 2(0\cdot(-1) - (-1)\cdot 2) = 4$.
$\det_3 = 2 \cdot 2 \cdot (-1) = -4$.
$\det(DF) = (-1)(-4) + (1)(4) + (-1)(-4) = 4 + 4 + 4 = 12 \neq 0$. ✓
By the (Inverse) Function Theorem, the solution $(1, 1, 1, 2)$ is locally isolated.